Remote Sensing Semantic Segmentation Quality Assessment Based on Vision Language Model

Huiying Shi, Zhihong Tan, Zhihan Zhang, Hongchen Wei, Yaosi Hu, Yingxue Zhang, and Zhenzhong Chen, *Senior Member, IEEE*

***Abstract*— Variations in scene complexity and image quality across remote sensing images lead to inconsistent performance when applying pretrained semantic segmentation models. To ensure quality control for downstream tasks, quality assessment on interpretation results becomes a challenging task, primarily due to the limited availability of reference annotations. To address this issue, we propose RS-SQA, a remote sensing semantic segmentation quality assessment model based on the vision language model (VLM) for images without ground truth. RS-SQA uses a dual-branch architecture that leverages an RS VLM for semantic understanding and utilizes intermediate features from segmentation methods to extract implicit information. We fuse these two feature streams via a semantic-guided module, yielding accurate quality predictions. Specifically, we introduce CLIP-RS, a large-scale pretrained VLM trained on 10 M remote sensing images with purified captions. Feature visualizations confirm that CLIP-RS can effectively differentiate between various levels of segmentation quality. To facilitate the development of RS semantic segmentation quality assessment, we present RS-SQED, a new dedicated dataset sampled from four major RS semantic segmentation datasets and annotated with segmentation accuracy scores from eight representative segmentation methods. Experiments on the established dataset demonstrate that RS-SQA outperforms state-of-the-art (SOTA) quality assessment models and aids in selecting the most suitable pretrained models for remote sensing semantic segmentation by comparing predicted accuracies across different models.**

***Index Terms*— Image quality assessment (IQA), remote sensing, remote sensing semantic segmentation quality assessment (RS-SQA), vision language model (VLM).**

I. Introduction

REMOTE sensing imagery (RSI), distinguished by its high resolution and extensive spatial coverage, has become a cornerstone data source in various applications such as land-cover classification [1], environmental monitoring [2], and disaster management [3]. Semantic segmentation, which assigns labels to individual pixels, is crucial for many RS interpretation practices. However, manual annotation of large RSI volumes is prohibitively time-consuming and costly. Consequently, the lack of ground truth labels for RSIs presents a significant challenge for the development and deployment of segmentation models.

From the perspective of model development, semi-supervised or weakly supervised learning methods are proposed to reduce annotation costs and enhance the model's generalization capabilities [4]. While pseudo-label noise can negatively impact training, highlighting the need for uncertainty assessment of the segmented samples. Our approach, however, primarily focuses on addressing the challenge during model deployment, especially when multiple models are available as options. Once a pretrained semantic segmentation model is deployed in practice, assessing its real performance on unseen data samples becomes a necessary step. Segmentation precision varies across image instances due to the complex spectral characteristics and significant object differences of RSI. Moreover, different methods exhibit varying adaptability to the same sample, complicating optimal segmentation model selection. Therefore, in order to guarantee the accuracy and facilitate the efficiency of the segmentation process, effectively assessing the quality of semantic segmentation is a pressing issue: "How can semantic segmentation quality be effectively assessed without ground truth?"

Despite its importance, research on quality assessment for unlabeled RSI semantic segmentation is limited. Methods in geo-object-based image analysis (GEOBIA), a type of object-based segmentation method, assess multiscale segmentation results based on intraregion homogeneity and inter-region heterogeneity until optimal scale parameters are found [5]. In parallel, uncertainty assessment methods for land-cover mapping often utilize deep Bayesian learning [6] and maximum softmax probability [7], [8], [9] as a proxy for prediction confidence. Since the ultimate goal is to improve the segmentation model's performance, the aforementioned methods focus on the inconsistency of insufficient and sufficient training. These approaches cannot quantitatively describe direct segmentation accuracy or compare quality across different segmentation models. More related studies [10], [11] simulate the Kappa coefficient on artificially constructed, compression-degraded RSI datasets. Despite providing useful insights, these machine learning methods are limited to specific degradation types and struggle to capture complex, real-world quality variations due to RSI's spatial heterogeneity and spectral ambiguity. To our knowledge, there is a clear need

Received 25 January 2025; revised 4 June 2025; accepted 7 July 2025. Date of publication 15 July 2025; date of current version 23 July 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62036005. *(Corresponding author: Zhenzhong Chen.)*

Huiying Shi, Zhihong Tan, Zhihan Zhang, Hongchen Wei, and Zhenzhong Chen are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430072, China (e-mail: zzchen@ieee.org).

Yaosi Hu is with the Department of Computing, The Hong Kong Polytechnic University, Hong Kong, SAR, China (e-mail: yaosi.hu@polyu.edu.hk).

Yingxue Zhang is with the College of Artificial Intelligence, Tianjin University of Science and Technology, Tianjin 300457, China (e-mail: yxzhang@tust.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3588854

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

for a score-based unsupervised quality assessment directly applicable to unlabeled RSI semantic segmentation.

Recent advancements in vision language models (VLMs) and foundation models have shown significant promise in bridging the semantic gap in various remote sensing tasks [12], [13], as comprehensively reviewed by Li et al. [14]. For instance, the pretrained vision encoder in VLMs with high-level scene understanding has been proven to explore the scene label information in embedding space in RSI scene classification [15]. Furthermore, their formidable capabilities extend to low-level vision tasks requiring fine-grained localization, such as object detection [16], showcasing the strong generalizability. Aligning with this trend of unified vision language pretraining, we propose RS-SQA: a semantic segmentation quality assessment framework for RSI when there is no GT available.

The construction of our RS-SQA involves two steps: first, proposing CLIP-RS, a geographical VLM, and second, efficiently integrating its visual perception capabilities into a score-based segmentation assessment task. RS-SQA combines semantic features from CLIP-RS as strong geographical prior knowledge, with intermediate segmentation features extracted from the semantic segmentation models to form a dual-branch network. Specifically, the proposed CLIP-RS is a CLIP-based model contrastively trained on a 10-million geographical text–image dataset. To eliminate the impact of text noise in the original data, we adopt a semantic similarity-based text purification strategy for robust semantic understanding.

In addition, to support the training and evaluation of the model, we establish RS-SQED, a new dataset for RS-SQA. It is sampled from four commonly used remote sensing semantic segmentation datasets and labeled with the accuracy scores of eight representative deep learning-based remote sensing semantic segmentation methods. The results on the established dataset demonstrate that our method achieves comprehensive state-of-the-art (SOTA) performance, surpassing existing quality evaluation methods.

Given that our primary goal is to facilitate the identification of the most appropriate model during deployment, evaluating only the quality evaluation metrics is not sufficient. Therefore, we further assess the top-1 accuracy, which helps to demonstrate the practical value of RS-SQA in enhancing the efficiency of accurate semantic segmentation in RSIs. The workflow of using RS-SQA for RSI interpretation is shown in Fig. 1.

The major contributions of our work are threefold.

1) RS-SQA, a dual-branch framework, is designed for remote sensing semantic segmentation quality assessment. It can simultaneously predict semantic segmentation accuracy and recommend the best-performing model from a pool of eight with a 0.7350 accuracy.
2) A 10-million-scale high-quality remote sensing image–text dataset is constructed through a novel data purification strategy based on semantic similarity. Leveraging this dataset, a pretrained remote sensing VLM CLIP-RS is proposed, enabling powerful geo-visual perception capability.

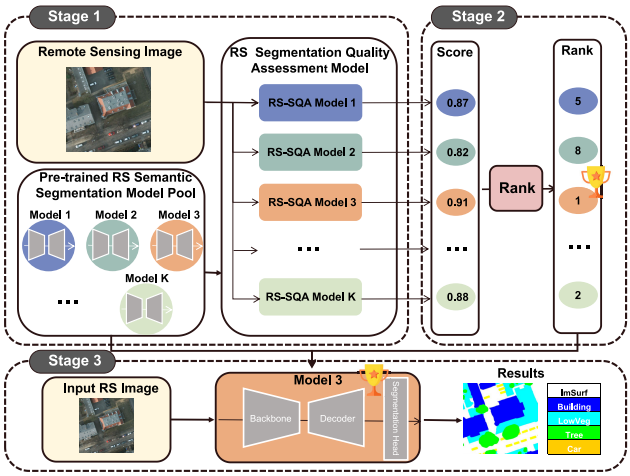


Fig. 1. Workflow for using the RS-SQA model to assist users in achieving optimal semantic segmentation. Stage 1: evaluate the semantic segmentation quality score for each available method. Stage 2: rank the methods based on their quality scores and select the top-performing one. Stage 3: apply the recommended model to segment the input image.

3) We have established the first large-scale remote sensing semantic segmentation quality evaluation dataset, RS-SQED, which covers diverse scenarios and is labeled with segmentation accuracy scores for eight different semantic segmentation methods.

II. Related Work

In this section, remote sensing semantic segmentation methods are reviewed. Notably, the application of the VLM for remote sensing semantic segmentation tasks is discussed. Furthermore, the semantic segmentation quality assessment metrics are also elaborated.

A. Remote Sensing Image Semantic Segmentation

Supervised deep learning has substantially advanced semantic segmentation in RSI. Fully convolutional networks (FCNs) [17] are widely used for pixel-level segmentation. To enhance the performance of FCNs on RSI, studies have incorporated techniques such as multiscale feature fusion [18] and the integration of auxiliary information (e.g., infrared images and digital surface models). Another widely adopted architecture is the encoder–decoder structure, exemplified by the renowned U-Net [19]. U-Net-like methods [20], [21], [22], [23] effectively combine deep and shallow features through skip connections, exhibiting excellent performance. To address the issue of easy loss of small-target information in RSI, multiscale feature fusion-based methods have also gained traction. Representative models like FPN [24] leverage the inherent pyramid structure of deep networks to combine features at different scales, preserving detailed information and achieving robust results in small-target segmentation tasks [25], [26]. Inspired by the success of Transformer in natural language processing, Transformer-based segmentation models have also emerged recently. These methods directly divide the image into patches, feeding them into the Transformer module for segmentation, fully exploiting Transformer's global

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

modeling capability. Exemplary works like ResT [27] and Segmenter [28] have also demonstrated promising performance.

Recent work has embarked on exploring VLMs for RSI that can be fine-tuned for semantic segmentation. Prior works leverage the contrastive learning strategy of image–text pairs in RS semantic segmentation [29]. RingMo [30] is the first generative self-supervised RS foundation model, pretrained on two million RS images, achieving SOTA on four downstream tasks, including semantic segmentation. Cha et al. [31] introduce the first billion-scale foundation model in the RS field, which achieves the best performance on the Potsdam and LoveDA [32] datasets. For multispectral images, some foundation models have been proposed such as SkySense [33] and SepctralEarth [34]. Recently, HyperFree [35] trained on large-scale hyperspectral images can process semantic segmentation tasks with promptable or zero-shot manner. Moreover, more remote sensing multimodal foundation models [12], [36] demonstrate robust zero-shot performance on various RS tasks, e.g., image and region captioning, visual question answering, scene classification, and visually grounded conversations.

To exploit unlabeled data and reduce labeling effort, semi-supervised techniques have been proposed. Methods [7], [8], [37] based on self-training with pseudo-labels leverage an initial model to predict segmentation labels for unlabeled pixels and screen the high-confidence predictions through dynamic entropy assessment for iterative retraining. Consistency-regularized (CR) based models [38], [39], [40] are commonly organized as teacher–student frameworks. Following the mean teacher paradigm, a student network is trained to match the predictions of a slowly updated teacher network under varying input perturbations such as color noise [38] or pasting labels [41]. Graph-based approaches [39], [42] construct a graph over pixel neighborhoods to integrate spectral clustering, especially for hyperspectral images, serving as kernel-based methods. For instance, NSCKL [39], which combines spectral clustering and kernel learning in a semi-supervised framework, demonstrated SOTA performance in hyperspectral image classification benchmarks.

B. Remote Sensing Unsupervised Segmentation Quality Assessment Methods

In the domain of remote sensing, unsupervised methods for segmentation quality assessment have been comparatively underexplored. Classical unsupervised evaluation methods rely on statistical and mathematical ways to represent the consistency in segmentation results. These methods typically involve the scoring and ranking of several image segmentations based on predefined quality criteria that align with human perceptual judgments of what constitutes an effective segmentation. A commonly accepted definition of high-quality segmentation includes: 1) regions that are internally consistent and homogeneous and 2) regions that exhibit clear dissimilarity from their neighboring segments [43]. Consequently, the majority of unsupervised evaluation approaches rely on calculating intrasegment and intersegment heterogeneity measures for each segment, for instance, Espindola et al. [5] introduced a method that quantified intrasegment homogeneity via weighted variance and intersegment heterogeneity through Global Moran's I, both computed from the near-infrared (NIR) band. Building upon similar principles, Kim et al. [44], Johnson and Xie [45] applied unweighted variance and Global Moran's I to land-cover segmentation. Additionally, hybrid evaluation frameworks combining unsupervised and supervised techniques have been proposed. Radoux and Defourny [46], for example, employed normalized post-segmentation standard deviation alongside border discrepancy metrics to assess segmentation quality in rural landscapes. These methods are all aimed at deriving the optimal scale for multiresolution segmentation (MRS) approach.

With the development of machine learning, some researchers predict segmentation accuracy such as Kappa coefficient [47] from images. Fractal [10] extracts local morphological dimensions and global multifractal structures, while Wei et al. [11] employ convolutional sparse coding to construct a regressor to assess segmentation quality. However, methods based on intraclass homogeneity struggle with the inherent dispersion in RSI. Machine learning approaches, constrained to ROI-based pixel classification, fail to generalize to deep learning-based segmentation quality assessment. Overall, unsupervised quality assessment for RS semantic segmentation still requires significant advancement.

III. Proposed Method

The proposed model, RS-SQA, comprises a high-level semantic feature extraction module, a segmentation feature extraction module, a feature fusion module, and a quality regression module. The framework of the proposed method is illustrated in Fig. 2. The details of each module are described in this section.

A. CLIP-RS: Vision Language Pretraining With Data Purification for Remote Sensing

Several studies have demonstrated the effectiveness of CLIP in the IQA [48] task due to its ability to achieve comprehensive cross-modality understanding by training on web-scale image–text datasets. The prominent VLM has demonstrated great promise to offer a broad scope of vision tasks. Specifically, CLIP comprises an image encoder and a text encoder, both jointly trained to establish a shared latent space for image and text through contrastive learning.

Given a batch of N paired images and texts $\{y_i, z_i\}_{i=1}^{N}$, CLIP extracts image features $\mathbf{U} = \{u(y_i)\}_{i=1}^{N}$ and text features $\mathbf{V} = \{v(z_i)\}_{i=1}^{N}$ with corresponding encoders. During pretraining, the model aims to maximize the cosine similarity of paired image and text features while minimizing the similarity of unmatched pairs. The contrastive learning objective can be formulated as

$$\mathcal{L}_{\text{image}} = -\mathbb{E}_{U_i \sim \mathbf{U}} \left[\log \frac{\exp(U_i^\top V_i/\alpha)}{\sum_{j=1}^{N} \exp(U_i^\top V_j/\alpha)} \right] \tag{1}$$

$$\mathcal{L}_{\text{text}} = -\mathbb{E}_{V_i \sim \mathbf{V}} \left[\log \frac{\exp(V_i^\top U_i/\alpha)}{\sum_{j=1}^{N} \exp(V_i^\top U_j/\alpha)} \right] \tag{2}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

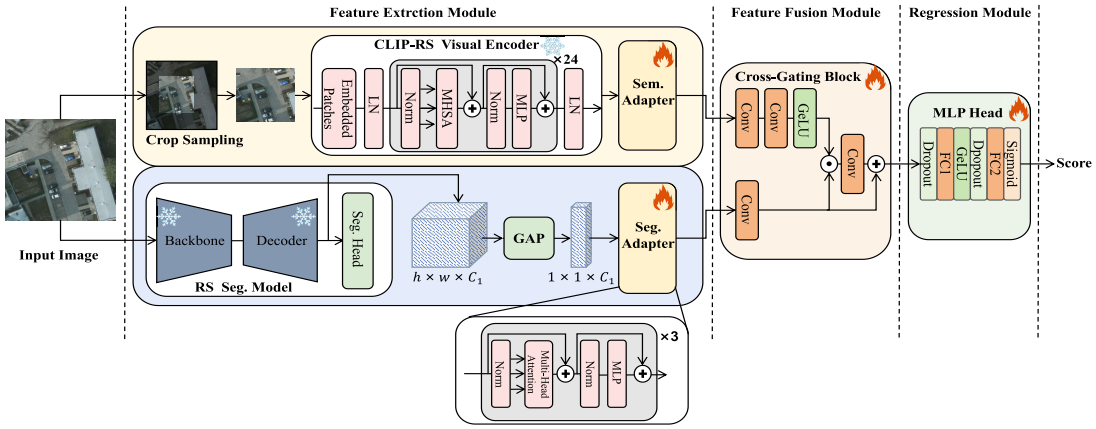


Fig. 2. Illustration of our framework. High-level semantic features are extracted from the CLIP-RS visual encoder, while deep segmentation features are obtained from the RS semantic segmentation model and simplified via average pooling. The features from both branches are fused using a cross-gating block and then input into a quality prediction head to generate the quality score. Seg.: segmentation. Sem.: semantic.

where U_i and V_i are the ith features of the batch and α is the temperature parameter. The final contrastive learning loss can be obtained by taking the average: $\mathcal{L} = (\mathcal{L}_{\text{image}} + \mathcal{L}_{\text{text}})/2$. With the contrastive learning strategy, CLIP can generate aligned cross-modality features and obtain strong image understanding. Based on this, we adopt the CLIP image encoder to extract semantic-sensitive features.

However, to effectively transfer CLIP to the remote sensing domain while maintaining its robust semantic perception capabilities, fine-tuning on the large-scale high-quality image–text data is essential. Existing datasets are far from sufficient to meet the requirements, for instance, the data volume of RemoteCLIP is several orders of magnitude less than that of CLIP. In light of this limitation, a large-scale, high-quality remote sensing image–text dataset comprising 10-million remote sensing images is constructed. This dataset is refined and cleaned using multimodal large language models (MLLMs) specialized in remote sensing.

The process of the construction of CLIP-RS is as follows.

1) Data Collection: Images are collected from globally open-source remote sensing image–text datasets. These datasets can be broadly categorized into two main types. The first type is exemplified by the RemoteCLIP dataset [12], which generates high-quality captions via ingenious structured generation rules, containing approximately 1.5 million images. The second type consists of 8.5 million images covering coarse semantic tags with unstructured information that might have a tenuous connection to geography and is notably heterogeneous in quality. Some of them carry a certain degree of semantic meaning and can offer assistance in understanding the related images such as " a satellite image of landuse of forest," while a portion is marred by noise and inaccuracies, for example, "Google Earth to photograph by Benjamin Grant." Despite the fact that the second type of captions enriches the data in terms of size and diversity, it also poses challenges since it causes hallucination problems in the pretrained model [49].

2) Data Filtering: To effectively exploit the dataset, a novel data filtering strategy is proposed to identify captions that need to be refined from the second type of captions, which is illustrated in Fig. 3. First, a high-quality semantic-aware CLIP model, denoted CLIP_{Sem}, is obtained through fine-tuning with 1.5 million high-quality image–text pairs based on the base CLIP model. Then, for the 8.5 million rough pairs, we input them into CLIP_{Sem} to extract the image features and text features. Referring to the cosine similarity loss in CLIP, the similarity score (SS) is calculated for each pair. Specifically, for a given image–text pair, the SS is defined as

$$\text{SS} = \frac{\mathbf{v}_I \cdot \mathbf{v}_T}{\|\mathbf{v}_I\| \|\mathbf{v}_T\|} \tag{3}$$

where $\mathbf{v}_I$ and $\mathbf{v}_T$ represent the embedding vectors extracted from visual and text spaces of CLIP_{Sem}, respectively. The cosine SS serves as a criterion for evaluating how well semantic tags match and their level of detail. Utilizing this SS, the bottom 20% of the 8.5 million data are identified as low-quality caption data, while the rest are categorized as high-quality caption data.

3) Data Refinement: The grounded remote sensing VLM, GeoChat [36], is utilized for refining and regenerating the low-quality captions. Due to the inherent domain disparity between remote sensing images and general images, MLLMs pretrained on remote sensing images yield favorable results in RS visual caption tasks. To guide GeoChat in generating concise and detailed descriptions that highlight the image's key visual and contextual features, a structured input–output interaction framework was employed. The process involves three key components.

1) *Instruction:* "Generate a brief description of the remote sensing image, highlighting key features such as the terrain, environment, layout, or other notable elements visible in the image."
2) *Metacaption:*"Description data of the image (insert the following data based on the actual image): [title]."
3) *Example:* "A satellite image of a coastal city with a network of roads, high-rise buildings, and a large harbor area."

We then recalculate the SS for the refined image–text pairs. If scores improve, replace original captions with

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

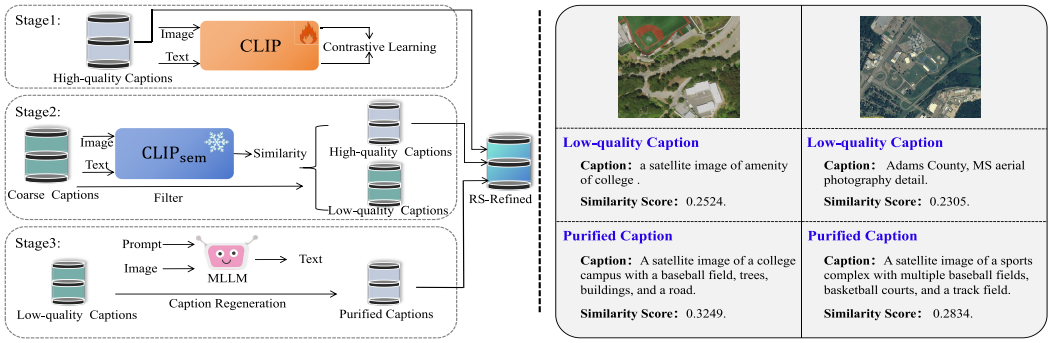


Fig. 3. Data purification process of the CLIP-RS dataset. (Left) Data purification workflow for CLIP-RS dataset. Stage 1: train CLIP to obtain CLIP_{Sem} with high-quality captions. Stage 2: use the pretrained CLIP_{Sem} to calculate image–text similarity. Stage 3: employ a remote sensing MLLM to regenerate captions for low-quality data. (Right) Examples of captioning results, showing initial low-quality image–text pairs and their corresponding purified captions.

newly generated ones; otherwise, keep the original captions. Fig. 3 illustrates the significant improvements in caption quality through the refinement process, showcasing comparisons between low-quality original captions and their refined GeoChat outputs.

4) Vision Language Pretraining: Based on a diverse collection of high-quality and purified image–text pairs, we construct the 10-million dataset and the CLIP-RS model is obtained by performing continual pretraining on the dataset starting from the CLIP model. This process equips the CLIP-RS with the ability of acquiring consistent semantic understanding of remote sensing images and encoding visual information.

B. RS-SQA Framework

1) Semantic Feature Extraction Module: The pretrained ViT-L-14 visual encoder from CLIP-RS is employed to capture the essential visual cues and object relationships with domain-specific prior knowledge.

Given an image I, it is fed into the visual encoder of blueCLIP-RS to acquire the CLS tokens, which is typically considered as the high-level semantic feature vector V_{Sem}

$$V_{\text{Sem}} = \text{Encoder}_{\text{CLIP-RS}}(I). \tag{4}$$

Since the CLIP-RS visual encoder has not undergone specific training for semantic segmentation quality tasks, it is mainly related to the abstractness of quality-aware features, potentially leading to a loss of semantic information during the encoding process into latent space. To address this issue, we propose a projection module named semantic adapter that interprets semantic features into a quality space. This module consists of three Vision Transformer (ViT) blocks, each incorporating a multihead self-attention (MHSA) mechanism and a feedforward MLP, as shown in Fig. 2. Specifically, the MHSA uses eight attention heads, and the MLP is composed of two linear layers, with a GELU activation function in between and followed by dropout. The process of extracting the semantic feature is as follows:

$$F_{\text{Sem}} = A_{\text{Sem}}(V_{\text{Sem}}) \tag{5}$$

where F_{Sem} denotes semantic features extracted from the semantic branch and A_{Sem} denotes the semantic adapter.

2) Segmentation Feature Extraction Module: Although CLIP excels at capturing global features, it primarily focuses on high-level semantic information and cannot make full use of low-level features such as texture, blur, color, and brightness, which are crucial for pixel-level tasks. Since semantic segmentation quality assessment is inherently a low-level visual task that relies heavily on shallow visual features, integrating segmentation-specific features can better capture these fine-grained details.

Furthermore, different semantic segmentation methods employ various structures for feature extraction, leading to differences in the areas of focus within the segmentation feature heatmaps, as shown in Fig. 4. These variations affect the segmentation performance across different remote sensing images. For instance, methods like UNetFormer [20] may prioritize capturing contextual information effectively, leading to smoother transitions between classes in the feature maps. In contrast, architectures such as MANet [22] and DC-Swin [23] may emphasize multiscale feature representation, which allows them to better delineate fine details in complex scenes. The observed differences in feature extraction can significantly impact segmentation performance across various remote sensing datasets. For example, in the ISPRS Potsdam dataset, the ability of BANet [27] to capture both local textures and global structures enhances its performance, while A2FPN [24] may struggle with finer details due to its focus on broader features. Consequently, the semantic segmentation features extracted from the target remote sensing segmentation method to identify the differences between different model architectures for understanding images.

Given an image I, it is fed into the pretrained segmentation model to extract the segmentation feature before the layer preceding the classifier

$$M_{\text{Seg}} = \text{Encoder}_{\text{Seg}}(I) \tag{6}$$

where $\text{Encoder}_{\text{Seg}}$ denotes the semantic segmentation model without segmentation classification head and M_{Seg} denotes the segmentation feature map. Subsequently, spatial global average pooling (GAP) is applied on each semantic segmentation feature map, condensing the dimensions from $h \times w \times c_1$ to $1 \times 1 \times c_1$

$$V_{\text{Seg}} = \text{GAP}(M_{\text{Seg}}) \tag{7}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

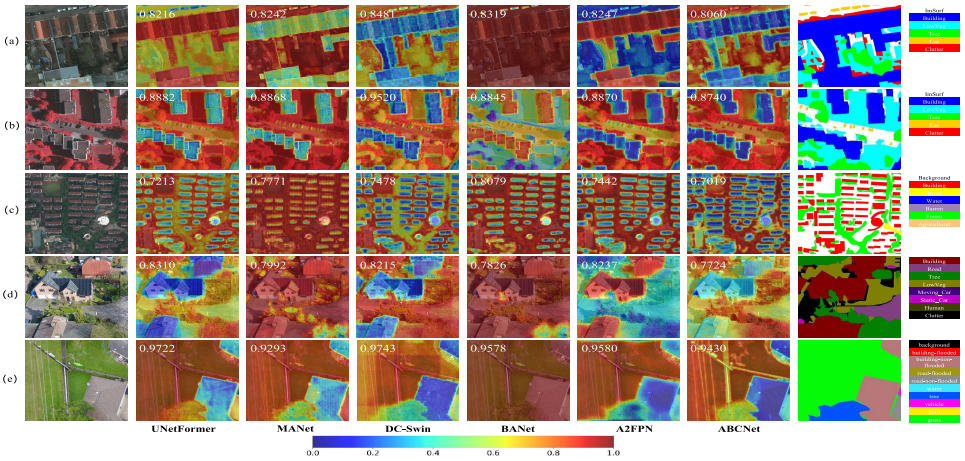


Fig. 4. Representative visualizations of feature heatmaps on remote sensing semantic segmentation datasets. Raw images, the feature maps extracted from different methods, and the ground truth labels are from left to right. Samples from (a)–(e) ISPRS Potsdam, ISPRS Vaihingen, LoveDA, UAVid, and FloodNet datasets, respectively. Features are overlaid on the raw image with 50% opacity, and the segmentation accuracy is indicated in the top-left corner.

where V_{Seg} refers to the average semantic segmentation feature vector.

To bridge the domain gap between semantic segmentation and quality assessment, a segmentation adapter is utilized to project the segmentation feature before feature fusion. Note that the segmentation adapter uses the exact same architecture as the semantic adapter—three ViT blocks—but is instantiated to accept the c_1 dimensional segmentation vector V_{Seg} rather than the CLS-token dimension of V_{Sem}

$$F_{\text{Seg}} = A_{\text{Seg}}(V_{\text{Seg}}) \tag{8}$$

where A_{Seg} represents the segmentation adapter and F_{Seg} denotes the feature extracted from the segmentation feature extraction module.

3) Feature Fusion Module: The CLIP-RS visual encoder demonstrates robust capabilities in representing image semantics through the vision language contrastive pretraining. Its global semantic perspective can effectively complement the segmentation features, which are more focused on local structures. Therefore, the potential correlation between the two branches requires effective integration. Here, a feature fusion block is utilized to harness the semantic feature to modulate the other branch that focuses on learning segmentation details. The novel channel modulating block is modified from the cross-gating block [50], and dubbed simple cross-gating block (SCGB), for feature fusion between semantic-aware and quality-aware feature pairs.

The SCGB operates on two input tensors, F_{Seg} and F_{Sem}, where F_{Seg} originates from the segmentation branch, and F_{Sem} represents features from the CLIP-based semantic branch. Through input channel projections, the projected CLIP features are fed to a gating pathway to yield the gating weights, which are then multiplied by the features from the other branch. Finally, a residual connection is adopted. The synergistic integration of semantic and segmentation features F_{Fusion} is formulated as

$$W'_{\text{Sem}} = \text{GELU}(W_{\text{Sem}} F_{\text{Sem}}) \tag{9}$$

$$F'_{\text{Seg}} = W_{\text{Seg}} F_{\text{Seg}} \tag{10}$$

$$F_{\text{Fusion}} = W_{\text{Fusion}}\left(W'_{\text{Sem}} \odot F'_{\text{Seg}}\right) + F'_{\text{Seg}} \tag{11}$$

where W_{Sem}, W_{Seg}, and W_{Fusion} are learnable projection matrices for channel modulation; GELU represents the activation function; and W'_{Sem} denotes the gating weights created by semantic features to integrate spatial semantic attention into segmentation features.

4) Quality Regression Module: The output of the fusion module is passed through a multilayer perceptron (MLP) to predict quality scores, as illustrated in Fig. 4. The MLP consists of two fully connected (FC) layers, interleaved with a GELU activation function. To address overfitting, dropout layers with a dropout rate of 0.1 are strategically positioned between the input features and the first FC layer, as well as between the GELU activation and the second FC layer. This architecture enhances generalization by reducing reliance on individual features while balancing complexity and performance. Finally, a sigmoid activation function is employed to map the linear outputs to values, aligning seamlessly with the requirements of segmentation quality assessment tasks ranging from 0 to 1.

5) Loss Function: To optimize our model, the loss function used for the proposed model consists of two parts: the mean squared error (mse) loss function and rank loss. The mse loss measures the prediction accuracy of the model, which is defined as

$$L_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(S_i - Q_i)^2 \tag{12}$$

where S_i and Q_i are subjective scores and predicted scores, respectively.

To further refine the model's ability to differentiate the distribution of segmentation quality scores, we introduce the Kullback—Leibler (KL) divergence loss function. This addition enhances the model's regression capability by penalizing differences between the predicted and true probability distributions. The KL divergence is defined as

$$L_{\text{KL}} = D_{\text{KL}}(P \| Q) = \sum_i P(i) \log \frac{P(i)}{Q(i)} \tag{13}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

TABLE I
OVERVIEW OF THE DATASET DISTRIBUTION

Database	Size	Count	Classes
ISPRS	1024×1024	902	ImSurf, Building, LowVeg, Tree, Car, Clutter
LoveDA	1024×1024	1669	Background, Building, Road, Water, Barren, Forest, Agricultural
UAVid	1024×1024	560	Building, Road, Tree, LowVeg, Moving-Car, Static-Car, Human, Clutter
FloodNet	1024×1024	3196	Background, Building, Road, Water, Tree, Vehicle, Pool, Grass

where P is the true probability distribution and Q is the approximate probability distribution of quality score.

The overall loss function is a weighted sum of the KL divergence loss and the mse loss function

$$L = L_{\mathrm{MSE}} + \alpha L_{\mathrm{KL}} \tag{14}$$

where α is the arithmetic weight of L_{KL} which is set to 1. This weighted combination allows the model to balance prediction accuracy with distributional alignment, leading to improved performance.

IV. EXPERIMENTAL RESULTS

In this section, we first describe the process of constructing the remote sensing semantic segmentation quality evaluation dataset. The experimental results and analysis present the accuracy of RS-SQA in recommending the optimal semantic segmentation method.

A. Database

The RS-SQED is constructed following the three-step construction scheme outlined in [10]. The first step is to select the source images from four commonly used semantic segmentation datasets. The second step involves the selection and retraining of the semantic segmentation model. We collected eight representative RS semantic segmentation methods as candidates and retrained them on each dataset. The third step is the standardization of the semantic segmentation quality score. Each of these steps is elaborated in Sections IV-A1– IV-A3.

1) Image Collection: The source images are primarily collected from four public remote sensing semantic segmentation datasets: ISPRS Vaihingen and Potsdam, LoveDA [32], UAVid [53], and FloodNet [54]. We selected images from these datasets for two main reasons: 1) these datasets have inherently considered the diversity of semantic scenes and spatial relationship between different objects and 2) these datasets possess segmentation annotations that are generally accepted.

1) The ISPRS Vaihingen and Potsdam (ISPRS) dataset is released by ISPRS Commission WG II/4. The Vaihingen dataset contains 33 VFR images, with an average size of 2494 × 2064 pixels and a ground sampling distance (GSD) of 9 cm. Only the TOP image tiles are used in training and test. The Potsdam dataset contains 38 images, whose average size is 6000 × 6000 pixels, and the resolution is 0.5 m. We utilize RGB channel images.
2) The LoveDA dataset comprises 5987 high-resolution images containing real urban and rural remote sensing images, each measuring 1024 × 1024 pixels with a GSD of 30 cm.
3) The UAVid dataset, as a fine-resolution uncrewed aerial vehicle (UAV) semantic segmentation dataset, designed for large-scale urban street scenes, consists of 300 high-resolution images of 3840 × 2160 pixels.
4) The FloodNet dataset is a high-resolution dataset aimed at post-flood scene understanding, comprising 1676 images with a GSD of approximately 1.5 cm. In order to maintain consistency with the water scene content in other datasets, images with the classification category "flooded" are not used.

To assess semantic segmentation quality without relying on annotations, we determined sampling proportions for each dataset based on their official divisions. Specifically, official test sets were utilized for ISPRS and FloodNet. For LoveDA and UAVid, where test set ground truth was unavailable during RS-SQED construction, validation sets were employed instead. The remaining portions of these datasets were used for retraining segmentation models.

The images for quality assessment mentioned above are divided into nonoverlapping patches and split into a training and testing set with an 8:2 ratio. Subsequently, they are cropped into 1024 × 1024 patches. The overview of the RS-SQED distribution is reported in Table I.

Some statistics of the RS-SQED dataset are analyzed to improve the universality of the dataset. As shown in Fig. 5, the number of annotated instances and pixels has been counted. We divide the original label categories of ISPRS, UAVid, LoveDA, and FloodNet datasets into seven categories according to natural vegetation, artificial surface, special surface, water, vehicle, human, and void for clear classification. The dataset contains a large number of objects and diverse categories, though with slight imbalance. Buildings and natural vegetation are more prevalent, as the dataset includes extensive sampling from FloodNet dominated by vegetation and UAVid, which has more buildings due to urban scene sampling. Conversely, the human and vehicle classes have relatively low pixel counts, primarily due to the smaller labeled area for human instances compared to buildings.

2) Semantic Segmentation Method Training and Inference: We explore a range of advanced methodologies for semantic segmentation in RSI, with a primary focus on deep learning-based models. These models, which typically follow an encoder–decoder architecture, include both Transformer-based and CNN-based approaches. It should be noted that our study does not include SOTA methods such as Deeplabv3+ [18], PSPNet [55], and S-RS-FCN [17].

We select eight methods based on diverse model structures to ensure a comprehensive evaluation. These include U-Net-like architectures (e.g., UNetFormer [20], AerialFormer [21], MANet [22], and DC-Swin [23]) and UperNet-like frameworks (e.g., UperNet (RSP-ViTAEv2-S) [52]). Additionally, we incorporate methods built with bilateral network frame-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

TABLE II
MEAN OA RESULTS OF THE RETRAINED SEMANTIC SEGMENTATION METHOD ON RS-SQED

Model	Time & Venue	Encoder	Decoder	Overall Accuracy (OA)				
				Potsdam	Vaihingen	LoveDA	UAVid	FloodNet
BANet [27]	2021 Remote Sens.	ResT-Lite	CNN	0.8864	0.8728	0.6715	0.8802	0.9253
ABCNet [52]	2021 ISPRS	CNN	CNN	0.8547	0.8598	0.5131	0.8348	0.9020
MANet [22]	2022 TGRS	ResNet50	CNN	0.8872	0.8787	0.6921	0.8859	0.8906
A2FPN [24]	2022 IJRS	CNN	CNN	0.8835	0.8715	0.6918	0.8790	0.9157
UperNet(RSP-ViTAEv2-S) [52]	2022 TGRS	RSP-ViTAEv2-S	CNN	0.8582	0.8758	0.6865	0.8941	0.9353
UNetFormer [20]	2022 ISPRS	ResNet18	Transformer	0.8840	0.8787	0.6713	0.8830	0.9387
DC-Swin [23]	2022 GRSL	Swin Transformer-S	CNN	0.8907	0.8802	0.7046	0.8982	0.9449
AerialFormer [21]	2024 Remote Sens.	Transformer	CNN	0.8764	0.8760	0.6992	0.8395	0.9123

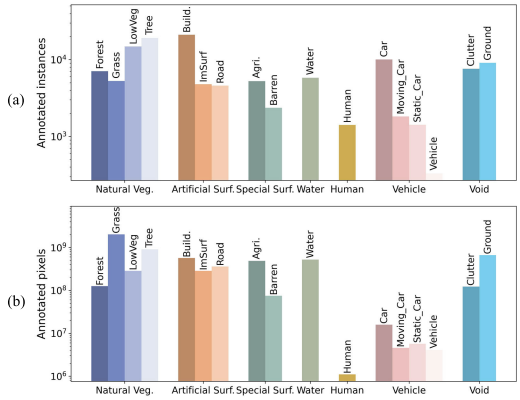


Fig. 5. Statistics for the pixels and objects in the RS-SQED dataset. (a) Histogram of the number of instances for each class. (b) Histogram of the number of pixels for each class. Buil.: building, Agri.: agriculture, Natural Veg.: natural vegetation, Artificial Surf.: artificial surface, and Special Surf.: special surface.

works (e.g., ABCNet [51] and BANet [27]) and FCN-like designs (e.g., A2FPN [24]) to ensure diversity in model design. Table II presents a summary of different models.

To avoid data leakage, theremaining portions of source datasets, which do not overlap with RS-SQED, are used for retraining the semantic segmentation methods. Each method is trained following this data partition, resulting in a total of 7×5 models, which are subsequently used to perform semantic segmentation in RS-SQED.

3) Label Construction: We adopt the overall accuracy (OA) of segmentation as the standardized metric for evaluating semantic segmentation quality as the study [56] recommended. OA is a commonly used overall indicator to measure segmentation accuracy, which denotes the percentage of correctly classified samples. Since RS-SQED is a hybrid dataset with diverse and imbalanced semantic categories, we use OA as a general score. The calculation of OA based on the confusion matrix is defined as

$$\mathrm{OA} = \frac{1}{N}\sum_{i=1}^{n} x_{ii} \tag{15}$$

where n is the sum of columns in the confusion matrix, which denotes the total number of categories; N is the total number of samples in the ground truth; and x_{ii} is the diagonal element of the confusion matrix, which represents the number of correctly classified samples. The mean OA results of the retrained semantic segmentation method on RS-SQED are shown in Table II. It is worth noting that the OA value is computed individually for each image in the dataset.

B. Experimental Setup

1) Implementation Details: Pretraining Settings: We train CLIP-RS on a single-node $7 \times$ NVIDIA GeForce RTX 4060 Ti machine. We initialize the CLIP/ViT-L-14 with the OpenAI model weights based on the performance of the initial model weights. The learning rate is set to $3e^{-9}$, the optimizer is AdamW, and the corresponding batch size is set to 7×16. The scheduler is cosine, with a 1000-step warmup, and the training epoch is 1. Other training details follow SkyScript [13].

RS-SQA Training Settings: We train RS-SQA on an NVIDIA TITAN XP machine. The AdamW optimizer is employed to optimize the model parameters, with the learning rate set to $1e^{-4}$. To ensure effective training, a two-stage learning rate scheduling strategy is adopted: lr_scheduler.LinearLR for warm-up and lr_scheduler.StepLR for formal training.

RS Segmentation Methods Training Settings: We retrain the candidate models on an NVIDIA TITAN XP machine. The semantic segmentation model's training procedure adheres to the learning rate and optimizer configurations specified in the open-source code. As we are not comparing different segmentation methods to discuss their performance, we set the batch size to 2 during training, such that the results from different methods could share the same configuration.

CLIP-RS DownStream Task Training Settings: For the downstream tasks of CLIP-RS, training is conducted on four NVIDIA GeForce RTX 3090 GPUs. Following RingMo and Skysense [30], [33], we use the UperNet as the segmentation framework. The AdamW optimizer is used. The LR and weight decay are set to $6e^{-5}$ and 0.01, respectively. The polynomial decay rule is applied with a power of 1.0 and a minimal LR of 0. The linear warming-up strategy is adopted at the first 1500 iterations, with a ratio of $1e^{-6}$. For the iSAID [57] dataset, the batch size and iterations are set to 2 and 160k, respectively.

2) Evaluation Metrics: We evaluate the capabilities of SQA using four commonly used metrics in image quality assessment (IQA), including Pearson's linear correlation coefficient (PLCC), Spearman's rank-order correlation coefficient (SROCC), root-mean-square error (RMSE), and Kendall

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

rank-order correlation coefficient (KROCC). The metrics are defined as follows:

$$\text{PLCC} = \frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{N}(x_i - \bar{x})^2 \sum_{i=1}^{N}(y_i - \bar{y})^2}} \tag{16}$$

$$\text{SROCC} = 1 - \frac{6\sum_{i=1}^{N} d_i^2}{N(N^2 - 1)} \tag{17}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - y_i)^2} \tag{18}$$

$$\text{KROCC} = \frac{2(P - Q)}{N(N - 1)} \tag{19}$$

where x_i and y_i represent the predicted and true scores, respectively, for each data point; $\bar{x}$ and $\bar{y}$ represent the mean values of x_i and y_i, respectively; d_i represents the difference between the ranks of each pair (x_i, y_i); N represents the total number of data points; and P and Q represent the number of concordant pairs and discordant pairs, respectively.

3) Compared Methods: Open-source unsupervised semantic SQA methods for remote sensing are scarce, making direct comparisons challenging. Therefore, we evaluate RS-SQA against several no-reference IQA (NR-IQA) algorithms. The comparison includes classical metrics such as BLIINDS-II [58] and BRISQUE [59], as well as the SOTA deep learning-based methods, including Topiq [60], AGAIQA [61], DEIQT [62], MANIQA [63], and HyperIQA [64]. Additionally, we incorporate a no-reference method [10] specifically designed for predicting segmentation accuracy in remote sensing images. We choose BLIINDS-II [58] and BRISQUE [59] to explore the effectiveness of subjective quality assessment based on the human visual system (HVS). We also select some representative deep-learning NR-IQA methods [61], [62], [63], [64] to validate the adaptability of data-driven methods in the task of semantic segmentation quality assessment. The compared methods are retrained on the RS-SQED.

C. Performance and Analysis of RS-SQA on the RS-SQED Dataset

We trained a quality assessment model for each semantic segmentation method using the RS-SQA framework. To enhance the model's robustness in assessing semantic segmentation quality across diverse scenarios, we employed a mixed dataset for training and conducted comprehensive evaluations on the test set.

The scatter plots between the predicted OA versus their ground truth under eight representative semantic segmentation methods are shown in Fig. 6. The horizontal and vertical axes denote the predicted OA and the ground truth, respectively, with each data point corresponding to an image. From the figure, it is observed that there is a strong linear correlation between the predicted performance and the true labels, which is consistent across both the mixed test set and individual semantic segmentation datasets for all eight methods. Additionally, in the mixed test set, the data points are uniformly and densely clustered, suggesting that the proposed method effectively predicts the semantic segmentation accuracy of remote sensing images. Among the individual test sets, the UAVid dataset exhibits the best prediction performance, with data points closely aligned along the fit line. This is likely due to the high ground sampling resolution and precise segmentation annotations, coupled with minimal content differences between the training and test sets of UAVid, which effectively reduce the impact of label uncertainty on the alignment between predicted values and true labels. Additionally, the strong prior knowledge of bird's-eye view scenes, embedded within CLIP-RS, allows RS-SQA to more effectively extract and comprehend the semantic features in UAVid images. Second, the FloodNet dataset exhibits a more pronounced long-tail effect, as most patches belong to one discriminative class. This leads to exceptionally high segmentation accuracy, with many instances achieving 100%. On the ISPRS dataset, the data points exhibit a significant deviation from the fit line. This phenomenon can be attributed to several factors. First, these datasets have a limited quantity of data points and constrained scene diversity, leading to overfitting and impacting generalization. Second, the ISPRS datasets generally feature high segmentation accuracy due to structured scenes (e.g., buildings and roads) and annotation biases toward dominant urban structures. This consistently high baseline accuracy makes discerning subtle quality variations challenging based on scene semantics. Finally, a critical factor is the spectral disparity: Vaihingen includes an NIR channel, creating tonal differences that can render semantic features less pronounced or distinct compared to the RGB spectral range our CLIP-RS backbone is pretrained on, thereby giving rise to inferior performance.

D. Performance Comparison of Different Quality Assessment Metrics

To quantitatively assess the effectiveness of our method, we present PLCC, SROCC, RMSE, and KROCC for various comparison methods on the RS-SQED dataset, as tabulated in Table III. It is evident that our method significantly enhances the accuracy of semantic segmentation quality assessment for remote sensing images. Compared to existing methods, ours achieves superior performance not only in correlation metrics, such as PLCC and SROCC, but also in terms of RMSE, which is critical for accurately estimating segmentation performance without requiring ground truth labels. The poor performance of RS-SQA on the ISPRS dataset can be attributed to the overfitting problem, spectral differences, and annotated class imbalance, as detailed earlier in Section IV-C.

Traditional IOA methods, namely BLIINDS-II [58] and BRISQUE [59], exhibit limited performance as they are primarily designed to evaluate subjective visual quality in natural images relying on natural scene statistics (NSS) such as local contrast and gradient distributions. However, remote sensing scenes prioritize geometric and semantic fidelity over fine-grained quality, rendering these features less discriminative for segmentation accuracy. For example, BRISQUE measures "naturalness" by modeling MSCN coefficients, which correlate poorly with semantic class errors in urban layouts. Deep learning-based methods, such as DEIQT [62], perform

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

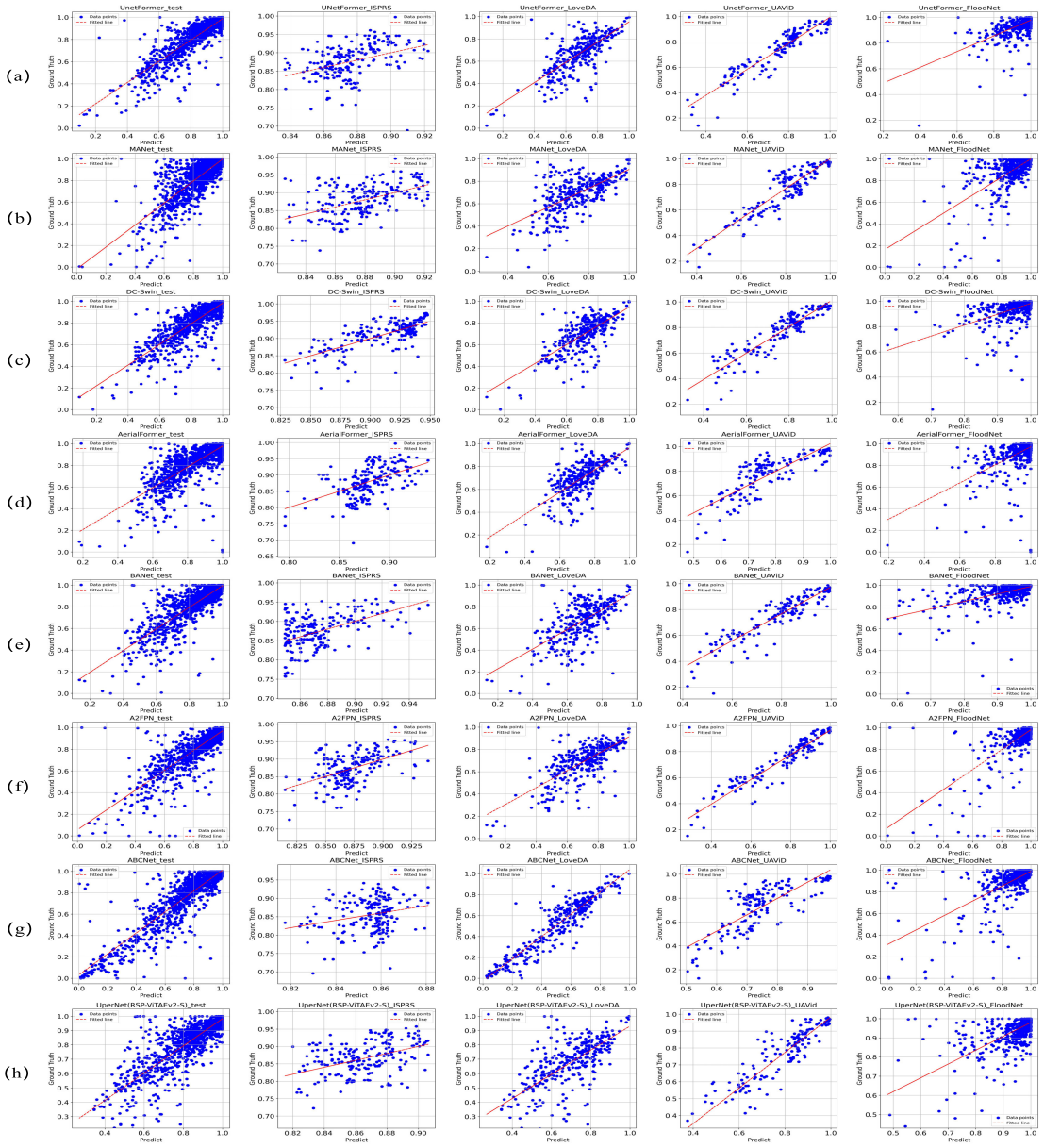


Fig. 6. Scatter plots between the predicted OA and the ground truth OA. The predicted OA is derived from models trained on ground truth segmented by UNetFormer [20], MANet [22], DC-Swin [23], AerialFormer [21], BANet [27], A2FPN [24], ABCNet [51], and UperNet (RSP-ViTAEv2-S) [52] [corresponding to (a)–(h)]. From left to right, the results correspond to RS-SQED, ISPRS Vaihingen and Potsdam, LoveDA, UAVid, and FloodNet datasets.

well on the UAVid and LoveDA datasets, likely due to the large training sets, which enable these methods to capture quality factors related to segmentation accuracy. However, the DEIQT's transformer backbone struggles to balance local details and global context. The lack of segmentation features also prevents the models from leveraging category-specific prior. The performance disparity between these NR-IQA methods and SQA methods reflects the fundamental divergence in evaluation paradigms between task-driven quality assessment and human perception-oriented IQA. Higher image clarity or structural distinctness does not always equate to better segmentation; resolution, scene complexity, and spectral/categorical consistency of ground objects also play a crucial role. There is also a special factor that affects the segmentation quality evaluation, the accuracy of the annotation.

E. Application on Recommending the Optimal Semantic Segmentation Method

To achieve the application-oriented goals that our model not only accurately predicts the segmentation accuracy of an image under a specific semantic segmentation method but also assists users in selecting the most accurate method among the numerous remote sensing segmentation options, we calculate the precision of recommending the top-one model among the candidate pool.

For image i, we use the quality assessment model trained with specific OA labels to predict the accuracy for each method, and the list of predicted scores is denoted as $S_i: \{s_{m1}, s_{m2}, s_{m3}, \ldots, s_{mk}\}$. The method corresponding to the highest score in S_i is the semantic segmentation method that our model evaluates as the most suitable for image i. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

TABLE III

PERFORMANCE COMPARISON OF THE RELATED MODELS ON RS-SQED. THE BOLD RESULTS AND THE UNDERLINED RESULTS INDICATE THE TOP AND THE SECOND PERFORMERS

Seg. Method	Model	RS-SQED				ISPRS		LoveDA		UAVid		FloodNet	
		PLCC	SROCC	RMSE	KROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC
BANet	BLIINDS-II [58]	0.0958	0.1584	0.1945	0.1053	0.2928	0.2330	0.0332	0.1881	0.4988	-0.4614	0.1070	0.1789
	BRISQUE [59]	0.3460	0.4239	0.1788	0.3027	0.2024	0.1509	0.3837	0.0920	0.2755	0.2068	0.1746	0.2126
	Fractal [10]	0.6631	0.7417	0.1602	0.6600	**0.6915**	**0.6475**	**0.6737**	**0.7114**	0.6931	0.7747	0.6446	**0.7572**
	MANIQA [63]	0.7296	0.7731	0.1236	0.5838	0.2356	0.1644	0.4366	0.3547	0.6580	0.5519	0.2870	0.4230
	HyperIQA [64]	0.6848	0.7079	0.1274	0.5328	0.4246	0.4639	0.5354	0.5109	0.4881	-0.2050	0.2896	0.3957
	DEIQT [62]	0.6829	0.7554	0.1871	0.6235	0.3611	0.3075	0.4741	0.4348	0.7919	0.7615	0.3210	0.5220
	AGAIQA [61]	0.7424	0.8101	0.1213	0.6254	0.2957	0.1845	0.4691	0.4178	0.6132	0.5605	0.4528	0.6125
	Topiq [60]	0.8074	**0.8761**	0.1102	**0.7077**	0.5655	0.6052	0.6381	0.6302	0.7926	0.7863	0.4403	0.7528
	Ours	**0.8448**	0.8670	**0.0935**	0.6963	0.4492	0.3547	0.6399	0.6019	**0.9168**	**0.9119**	**0.6657**	0.7148
ABCNet	BLIINDS-II [58]	0.1804	0.1991	0.2635	0.1318	0.1560	0.1230	0.0309	0.1795	0.4941	-0.4711	0.1735	0.1978
	BRISQUE [59]	0.3664	0.4074	0.2402	0.0449	0.1477	0.0449	0.2809	0.1196	0.2513	0.1778	0.1864	0.1916
	Fractal [10]	0.6708	0.8052	0.2272	0.6402	0.4636	**0.5014**	0.6689	0.6757	0.7284	0.8079	0.6522	0.7281
	MANIQA [63]	0.8037	0.7892	0.1540	0.6140	0.1373	0.0892	0.7799	0.7579	0.5419	0.4748	0.4329	0.5514
	HyperIQA [64]	0.7024	0.7779	0.1873	0.6028	0.2439	0.1764	0.4257	0.4914	0.2697	-0.1931	0.3259	0.5894
	DEIQT [62]	0.8351	0.8070	0.1212	0.6796	0.3011	0.1880	0.8869	0.8678	0.7557	0.6987	0.5971	0.6430
	AGAIQA [61]	0.8268	0.8351	0.1407	0.6666	0.3056	0.2582	0.8191	0.7975	0.5534	0.5210	0.4577	0.6263
	Topiq [60]	0.8634	0.8771	0.1265	0.7212	0.2486	0.2395	0.8968	0.8703	0.8438	0.7960	0.5294	0.7364
	Ours	**0.9050**	**0.8985**	**0.1078**	**0.7478**	**0.4639**	0.4658	**0.9352**	**0.9117**	**0.8616**	**0.8127**	**0.7033**	**0.7720**
MANet	BLIINDS-II [58]	0.0977	0.1700	0.1906	0.1131	0.2175	0.1215	0.0172	0.2072	0.5262	-0.5004	0.1252	0.1574
	BRISQUE [59]	0.3134	0.3807	0.1785	0.2703	0.2871	0.2141	0.1149	0.0589	0.2950	0.2243	0.1601	0.1842
	Fractal [10]	0.6758	0.7435	0.1559	0.6576	**0.5288**	**0.4827**	0.6422	**0.6940**	0.7062	0.7968	0.6496	**0.7673**
	MANIQA [63]	0.6092	0.6895	0.1503	0.5156	0.2432	0.0711	0.4954	0.4316	0.4661	0.4211	0.3798	0.5134
	HyperIQA [64]	0.6273	0.7038	0.1391	0.5371	0.4130	0.4313	0.5937	0.5306	0.1774	0.0572	0.5041	0.5632
	DEIQT [62]	0.6445	0.6991	0.1260	0.5816	0.3764	0.3796	**0.6802**	0.6000	0.7684	0.7282	0.5394	0.6052
	AGAIQA [61]	0.6679	0.7380	0.1334	0.5618	0.3171	0.3262	0.4896	0.4455	0.5205	0.5340	0.5294	0.5576
	Topiq [60]	0.7182	0.8147	0.1237	0.6499	0.4984	0.5203	0.6778	0.6442	0.7739	0.7867	0.4927	0.7105
	Ours	**0.7932**	**0.8255**	**0.1098**	0.6551	0.5025	0.4302	0.6747	0.6322	**0.9324**	**0.8799**	**0.6575**	0.7365
A2FPN	BLIINDS-II [58]	0.1188	0.1703	0.1887	0.1131	0.1971	0.1489	0.0353	0.1953	0.5112	-0.4717	0.1371	0.1952
	BRISQUE [59]	0.3083	0.4071	0.1756	0.2911	0.3073	0.2337	0.0809	0.0844	0.3008	0.2089	0.1848	0.2115
	Fractal [10]	0.6608	0.7501	0.1550	0.6684	**0.5509**	0.4505	0.6538	0.7262	0.7249	0.8065	0.6515	0.7672
	MANIQA [63]	0.6425	0.7591	0.1413	0.5779	0.3456	0.2697	0.5583	0.4474	0.6356	0.6159	0.3400	0.5093
	HyperIQA[64]	0.5964	0.6966	0.1480	0.5243	0.3164	0.2552	0.4391	0.4463	0.4900	0.2453	0.3835	0.4167
	DEIQT [62]	0.7125	0.7738	0.1780	0.6436	0.4270	0.3424	0.6541	0.6375	0.8197	0.7953	0.4703	0.5540
	AGAIQA [61]	0.6760	0.8008	0.1374	0.6224	0.3830	0.4056	0.5129	0.4478	0.4795	0.4335	0.4513	0.6168
	Topiq [60]	0.7246	0.8391	0.1269	0.6829	0.5266	**0.5536**	0.7254	0.6592	0.8165	0.7793	0.3772	0.7091
	Ours	**0.8571**	**0.8970**	**0.0945**	**0.7392**	0.5450	0.5282	**0.7456**	0.7004	**0.9454**	**0.9313**	**0.7788**	**0.8207**
RSP-ViTAEv2-S	BLIINDS-II [58]	0.1463	0.1809	0.1675	0.1207	0.2348	0.2248	0.2023	0.2155	0.4949	-0.4671	0.1955	0.2425
	BRISQUE [59]	0.3799	0.4333	0.1530	0.3072	0.1706	0.0558	0.3310	0.0990	0.2823	0.1828	0.2152	0.2314
	Fractal [10]	0.6914	0.8186	0.1346	0.6601	**0.5849**	**0.5936**	0.6621	**0.7072**	0.7566	0.8465	0.6489	0.7471
	MANIQA[63]	0.7207	0.7872	0.1182	0.5969	0.2325	0.1581	0.3776	0.3667	0.6875	0.6688	0.3679	0.4967
	HyperIQA [64]	0.6629	0.7320	0.1282	0.5518	0.3756	0.3844	0.4299	0.4370	0.2022	0.1223	0.3614	0.5337
	DEIQT [62]	0.7826	0.8333	0.1067	0.6507	0.3607	0.2152	0.6139	0.5939	0.8115	0.7556	0.4600	0.6344
	AGAIQA [61]	0.7465	0.8217	0.1181	0.6317	0.2860	0.1830	0.4660	0.3865	0.6797	0.6934	0.4345	0.6132
	Topiq [60]	0.8093	0.8747	0.1053	0.7043	0.3651	0.3605	0.6220	0.6377	0.8389	0.7942	0.5162	0.7909
	Ours	**0.8535**	**0.8809**	**0.0895**	**0.7186**	0.3948	0.3629	**0.6924**	0.6706	**0.8818**	**0.8747**	**0.7279**	**0.7964**
UNetFormer	BLIINDS-II [58]	0.1686	0.1914	0.1735	0.1274	0.2507	0.2023	0.1293	0.2135	0.5055	-0.4746	0.1891	0.2207
	BRISQUE [59]	0.4047	0.4633	0.1556	0.3323	0.2809	0.2288	0.3444	0.0792	0.3025	0.2109	0.2362	0.2572
	Fractal [10]	0.6938	0.8316	0.1387	0.6709	**0.6902**	**0.6341**	0.6524	**0.7013**	0.7007	0.8036	**0.6643**	0.7660
	MANIQA [63]	0.7653	0.8101	0.1117	0.6176	0.2918	0.0982	0.4962	0.3721	0.5668	0.5575	0.3288	0.4977
	HyperIQA [64]	0.7344	0.7877	0.1158	0.6034	0.3799	0.4431	0.5498	0.4637	0.1055	0.0955	0.3389	0.4550
	DEIQT [62]	0.7720	0.8177	0.0982	0.6711	0.3781	0.3674	0.6890	0.6219	0.7638	0.7633	0.4693	0.6276
	AGAIQA [61]	0.7981	0.8469	0.1099	0.6640	0.2859	0.3096	0.5855	0.5282	0.5949	0.5389	0.4378	0.5984
	Topiq [60]	0.8239	0.8912	0.1004	0.7196	0.4148	0.4729	0.6308	0.5913	0.8153	0.7783	0.5038	0.7686
	Ours	**0.8955**	**0.9105**	**0.0752**	**0.7539**	0.6329	0.5635	**0.7670**	**0.7451**	**0.9592**	**0.9580**	0.6320	**0.8017**
DC-Swin	BLIINDS-II [58]	0.1922	0.2223	0.1583	0.1495	0.0755	0.0627	0.1497	0.2170	0.5052	-0.4781	0.2064	0.2411
	BRISQUE [59]	0.4010	0.4441	0.1432	0.3171	0.5071	0.4604	0.0742	0.0485	0.3513	0.2664	0.2120	0.2286
	Fractal [10]	0.7156	0.7506	0.1267	0.6826	0.7099	0.6814	0.6532	**0.7418**	0.6759	0.7810	**0.6729**	0.7748
	MANIQA [63]	0.7425	0.8013	0.1092	0.6075	0.5491	0.5092	0.4588	0.4002	0.6748	0.6051	0.3482	0.5162
	HyperIQA [64]	0.6944	0.7544	0.1165	0.5710	0.6268	0.4927	0.4594	0.4265	0.1449	0.0845	0.3562	0.3880
	DEIQT [62]	0.7664	0.8112	0.0946	0.6636	0.6325	0.5708	0.6445	0.5512	0.8123	0.7730	0.4763	0.6544
	AGAIQA [61]	0.7639	0.8233	0.1153	0.6420	0.5578	0.5372	0.5501	0.4144	0.6290	0.6548	0.3394	0.5811
	Topiq [60]	0.8305	0.8851	0.0912	0.7206	0.5922	0.5916	0.7282	0.6378	0.8855	0.8718	0.5073	0.7668
	Ours	**0.8749**	**0.9097**	**0.0781**	**0.7469**	**0.7108**	**0.6877**	**0.7213**	0.6642	**0.9180**	**0.9018**	0.6057	**0.8228**
AerialFormer	BLIINDS-II [58]	0.0888	0.1419	0.1868	0.0927	0.2034	0.1590	0.0936	0.1702	0.4968	-0.4722	0.0940	0.1356
	BRISQUE [59]	0.3393	0.4023	0.1723	0.2837	0.1977	0.1129	0.3338	0.0513	0.2974	0.2045	0.1913	0.2269
	Fractal [10]	0.6826	0.7474	0.1516	0.6626	**0.6049**	**0.6424**	**0.6751**	**0.7273**	0.7001	0.8065	**0.6557**	**0.7652**
	MANIQA [63]	0.6449	0.7389	0.1293	0.5596	0.2818	0.2358	0.3920	0.3347	0.7021	0.6771	0.3084	0.4994
	HyperIQA [64]	0.6298	0.6935	0.1295	0.5278	0.1983	0.1778	0.5200	0.5275	0.5138	0.5830	0.3220	0.4449
	DEIQT [62]	0.6560	0.7259	0.1163	0.5944	0.3363	0.3319	0.6063	0.5588	0.7451	0.7228	0.4486	0.5316
	AGAIQA [61]	0.6923	0.7707	0.1206	0.5856	0.2760	0.2114	0.4853	0.4541	0.6263	0.6036	0.4334	0.5412
	Topiq [60]	0.7454	0.8337	0.1131	0.6693	0.4486	0.4866	0.6485	0.5955	0.8212	0.7948	0.5008	0.7128
	Ours	**0.7951**	**0.8613**	**0.1029**	**0.6892**	0.4395	0.4392	0.6724	0.6461	**0.8605**	**0.8288**	0.5829	0.7497

precision is defined as

$$\text{Precision} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{L}(P_i \subseteq B_i) \tag{20}$$

where P_i and B_i are the list of predicted best methods and ground truth OA corresponding methods for image i, respectively. Given that multiple semantic segmentation methods may achieve the same segmentation accuracy for the same image, we define L as an indicator function, indicating that the predicted method list is a proper subset of the true optimal method list.

Specifically, Table IV presents the precision of recommending the top-performer across eight semantic segmentation techniques. P@1 denotes the accuracy of identifying the top-ranked model within the model pool, while P@3 represents the accuracy of the selected top-ranked model being among the top three models in the ground truth rankings.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
PRECISION COMPARISON OF THE BEST METHOD PREDICTION AMONG EIGHT MODELS. THE BOLD INDICATES THE TOP ONE PERFORMER

Model	Time & Venue	RS-SQED		ISPRS		LoveDA		UAVid		FloodNet	
		P@1	P@3	P@1	P@3	P@1	P@3	P@1	P@3	P@1	P@3
BLIINDS-II [58]	2012 TIP	0.4967	0.7172	0.2913	0.6359	0.4414	0.7417	0.5688	0.7125	0.5810	0.7445
BRISQUE [59]	2012 TIP	0.5714	0.7306	0.3447	0.6456	0.6787	0.7838	0.6625	0.7813	0.5748	0.7305
Fractal [10]	2019 TGRS	0.6706	0.7934	0.3816	0.6748	0.7310	0.8228	0.6643	0.8625	0.7440	0.8114
MANIQA [63]	2020 CVPR	0.6780	0.8031	0.4078	0.5741	0.7136	0.8318	0.5375	0.7125	0.7804	0.8364
HyperIQA [64]	2022 CVPR	0.6496	0.7496	0.2379	0.5534	0.7508	0.8559	0.5938	0.6625	0.7535	0.8019
DEIQT [62]	2023 AAAI	0.6854	0.8098	**0.4466**	**0.8155**	0.7447	0.8559	0.6875	0.7438	0.7414	0.8271
AGAIQA [61]	2024 TCSVT	0.6906	0.8120	0.4029	0.7184	0.7147	0.8288	0.6438	0.7563	0.7928	0.8598
Topiq [60]	2024 TIP	0.7024	0.8127	0.3835	0.6068	0.7458	0.8455	0.6534	0.8250	0.7975	0.8478
Random	-	0.1333	0.4296	0.1408	0.3301	0.1532	0.3544	0.1438	0.3750	0.1201	0.5211
Ours	-	**0.7350**	**0.8320**	0.4078	0.6408	**0.7518**	**0.8561**	**0.8250**	**0.8875**	**0.8209**	**0.8816**

TABLE V
CROSS-DATASET COMPARISON ON THE VDD DATASET

Seg. Method	Ours		Topiq		DEIQT	
	PLCC	SROCC	PLCC	SROCC	PLCC	SROCC
BANet	0.6727	0.6132	0.2802	0.1389	0.2101	-0.1376
ABCNet	0.4584	0.4654	0.4045	0.2774	0.1483	-0.1624
MANet	0.5701	0.5721	0.3137	0.1854	0.1054	0.0353
A2FPN	0.6562	0.6382	0.1159	0.1201	0.0738	0.0271
RSP-ViTAEv2-S	0.7046	0.6911	0.2214	0.1526	0.2380	0.2452
UNetFormer	0.6736	0.5895	0.1426	0.0846	0.1531	0.1163
DC-Swin	0.7045	0.7445	0.1445	-0.0942	0.1999	0.0918
AerialFormer	0.6517	0.6207	0.1681	0.1720	0.1484	0.1404

TABLE VI
FROZEN BACKBONE TUNING SEMANTIC SEGMENTATION ON THE iSAID DATASET

Model	Time & Venue	mIoU
CMID [65]	2023 TGRS	59.4
SatLas [66]	2023 ICCV	56.03
GFM [67]	2023 ICCV	60.86
Scale-MAE [68]	2023 ICCV	46.53
CLIP-RS (1.5M)	-	59.89
CLIP-RS (10M)	-	61.31(+2.4%)

Compared to Fractal [10] (the approach most conceptually aligned with ours), our method demonstrates superior performance across all datasets except the ISPRS dataset. The performance gap on ISPRS can be attributed to the poor prediction accuracy of RS-SQA on the semantic segmentation quality score, as shown in Table III. This also highlights the difficulty of RS-SQA, due to the vast field of view and intricate textures of RSI.

When benchmarked against more sophisticated deep learning-based IQA methods, such as Topiq and AGAIQA, Topiq achieves results comparable to ours. However, our method surpasses it by a margin of 4.6%, further confirming its efficacy. This finding highlights the capability of RS-SQA to effectively evaluate the unique scene semantics inherent in RSI grounded in powerful pretraining and segmentation features. With an accuracy of 0.7350, RS-SQA reliably recommends the optimal semantic segmentation approach for RSIs.

F. Cross-Dataset Generalizability

We also run cross-dataset tests on the VDD dataset [69] to verify the generalizability of the proposed model. RS-SQA is trained on RS-SQED and then evaluated on VDD without any adjustments to its parameters. Given the domain gap between the VDD dataset and the pretrained weights, we map the VDD dataset's categories to RS-SQED's four data subsets. The mean segmentation accuracy across these four subsets is seen as the pseudo-ground truth.

From Table IV, we can see that DEIQT and Topiq are two competitive methods, so we use these methods as comparisons to verify the generalization of our model. The results are shown in Table V from which we can find that the proposed model shows good performance as a result of incorporating segmentation features. Therefore, we conclude that the proposed model is reasonable and has good generalizability.

G. Ablation Study of the Pretraining Scale on Semantic Segmentation

Considering factors such as spatial resolution, spectrum, and number of categories, we select one popular dataset to evaluate the effect of pretraining scales. To validate the superiority of the features learned through our pretraining, we carry out experiments using frozen backbone features on iSAID and report mIoU values of all categories. The results of the comparative method [65], [66], [67], [68] were adopted from [33], with consistent training settings used for a fair comparison. As shown in Table VI, our CLIP-RS achieves superior performance with the fixed backbone and the mIoU performance of the CLIP-RS model improves significantly as the training scale increases.

H. Ablation Study on Recommending the Optimal Semantic Segmentation Method

In this section, we carry out extensive ablation experiments to demonstrate the effectiveness of the model design as well as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

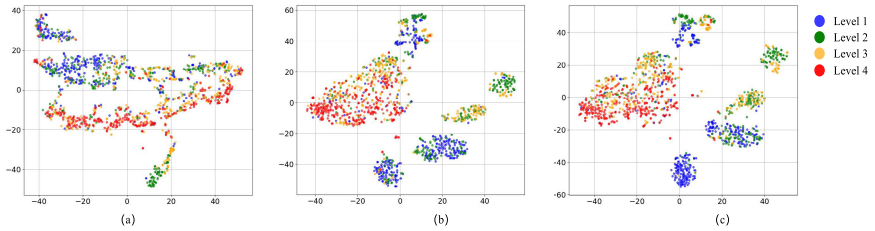


Fig. 7. t-SNE method visualization of the semantic features extracted from the visual encoder of different VLMs. The samples are labeled with UperNet (RSP-ViTAEv2-S) segmentation accuracy scores and categorized into four evenly distributed levels. (a) LIP. (b) RemoteCLIP. (c) CLIP-RS.

TABLE VII

ABLATION STUDY OF THE MODEL DESIGN ON THE RS-SQED DATASET. THE PRECISION OF PREDICTING THE BEST METHOD AMONG EIGHT MODELS. THE BOLD INDICATES THE TOP ONE PERFORMER

Model	Sem. E	Sem. A	Seg. E	Seg.A	SCGB	RS-SQED		ISPRS		LoveDA		UAVid		FloodNet	
						P@1	P@3	P@1	P@3	P@1	P@3	P@1	P@3	P@1	P@3
w/o adapter	✓		✓		✓	0.6304	0.7477	0.2796	0.5720	0.6658	0.7829	0.6163	0.7313	0.7380	0.8016
w/o SCGB (Concat.)	✓	✓	✓	✓		0.6203	0.7720	0.1796	0.5631	0.6607	0.7958	0.7688	0.8250	0.7134	0.8224
w/o SCGB (Atten.)	✓	✓	✓	✓		0.6336	0.7661	0.2379	0.4951	0.6456	0.8048	0.7375	0.8063	0.7383	0.8209
w/o seg.	✓	✓				0.6721	0.7802	0.2282	0.5437	0.7147	0.8198	0.6688	0.7750	0.8037	0.8520
w/o sem.			✓	✓		0.6773	0.8061	0.3641	**0.6505**	0.6607	0.7898	0.7500	0.8250	0.7788	0.8583
Ours	✓	✓	✓	✓	✓	**0.7350**	**0.8320**	**0.4078**	0.6408	**0.7518**	**0.8561**	**0.8250**	**0.8875**	**0.8209**	**0.8816**

TABLE VIII

ABLATION STUDY OF THE LOSS FUNCTION ON THE RS-SQED DATASET. THE PRECISION OF PREDICTING THE BEST METHOD AMONG EIGHT CANDIDATE METHODS

Strategy	RS-SQED	ISPRS	LoveDA	UAVid	FloodNet
w/o L_{MSE}	0.7195	0.3738	0.7267	0.8125	0.8146
$\alpha = 0$	0.7061	0.3544	0.7508	0.7438	0.7975
$\alpha = 0.3$	0.7343	0.4029	0.7117	0.8375	**0.8380**
$\alpha = 0.5$	0.7328	0.3883	**0.7748**	0.8375	0.8069
$\alpha = 0.7$	0.7306	**0.4223**	0.7237	**0.8438**	0.8162
$\alpha = 1$	**0.7350**	0.4078	0.7518	0.8250	0.8209

TABLE IX

ABLATION STUDY OF THE VISUAL ENCODER FROM DIFFERENT VLMS ON THE RS-SQED DATASET. THE PRECISION OF PREDICTING THE BEST METHOD AMONG EIGHT CANDIDATE METHODS

Model	RS-SQED	ISPRS	LoveDA	UAVid	FloodNet
RemoteCLIP [12]	0.6906	0.3252	0.7057	0.6125	0.8302
CLIP-RS (1.5M)	0.7291	0.3835	0.7387	0.8125	0.8255
CLIP-RS (10M)	0.7350	0.4078	0.7518	0.8250	0.8209

the large-scale pretrained VLM in the task of recommending the optimal semantic segmentation method.

1) Effectiveness of the Model Design: To evaluate the effectiveness of the model design, we conduct a set of experiments on the RS-SQED dataset and report the Precision metric IV-E. As shown in Table VII, directly fusing CLIP-RS features with segmentation features ("w/o adapter") degrades performance, confirming the adapter's importance in latent space alignment. Replacing SCGB with attention mechanisms or simple concatenation yields inferior results, underscoring its effective feature integration. Experiments with single-branch configurations reveal that segmentation characteristics ("w/o sem.") provide essential spatial details for quality assessment, while CLIP-RS features ("w/o seg.") offer complementary semantic information, jointly validating the architecture's effectiveness in cross-modal feature integration.

2) Effectiveness of the Loss Function: To further validate the effectiveness of the KL divergence joint loss function and investigate the network's sensitivity to the balancing weight parameters α, a series of ablation experiments are conducted. The results of these experiments are presented in Table VIII, where RS-SQA achieves the best performance on the RS-SQED when α is set to 1. The inclusion of L_{KL} contributes to an around 3.9% gain compared to the variant of "$\alpha = 0$," indicating the joint loss of L_{MSE} and L_{KL} exhibits substantial improvements.

3) Effectiveness of the VLM: To verify the effectiveness of large-scale high-quality pretraining, we compare CLIP-RS with RemoteCLIP [12], a representative VLM chosen for its strong performance, on the precision of recommending the best semantic segmentation method. Both CLIP-RS and RemoteCLIP [12] are based on the CLIP model, sharing identical structures that allow them to be directly interchangeable without modification to the input and output of other modules. As shown in Table IX, the CLIP-RS demonstrates significant 6.9% advances over RemoteCLIP [12]. Notably, the 1.5 M variant of CLIP-RS (trained on original high-quality captions, excluding purified captions) also outperforms RemoteCLIP, demonstrating the importance of well-organized datasets in facilitating semantic alignment. Furthermore, to verify the correlation between semantic information and segmentation accuracy, feature visualizations from the encoders of general CLIP, RemoteCLIP, and CLIP-RS are shown in Fig. 7. These visualization results confirm that CLIP-RS more clearly distinguishes categories related to segmentation quality, thereby enhancing the performance of segmentation quality assessment.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

V. Conclusion

In this article, we present RS-SQA, a novel RS-SQA framework. RS-SQA leverages a dual-branch design, integrating high-level semantic features from a large-scale pretrained VLM (CLIP-RS) with detailed segmentation features for comprehensive quality evaluation. To facilitate this field, we also establish RS-SQED, a meticulously crafted dataset from five common RS semantic segmentation datasets, annotated with accuracy scores from eight diverse methods. Experiments on RS-SQED demonstrate RS-SQA's superior performance over SOTA models and its effectiveness in recommending suitable segmentation methods, proving its value for efficient geospatial data analysis. Future work will extend the framework to broader RS tasks, addressing potential performance influences from data diversity and quality.

Acknowledgment

The numerical calculations in this article have been done on the supercomputing system in the Supercomputing Center, Wuhan University, Wuhan, China.

References

[1] M. Kampffmeyer, A.-B. Salberg, and R. Jenssen, "Semantic segmentation of small objects and modeling of uncertainty in urban remote sensing images using deep convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2016, pp. 680–688.

[2] J. Li, Y. Pei, S. Zhao, R. Xiao, X. Sang, and C. Zhang, "A review of remote sensing for environmental monitoring in China," *Remote Sens.*, vol. 12, no. 7, p. 1130, Apr. 2020.

[3] C. Van Westen, "Remote sensing for natural disaster management," *Int. Arch. Photogramm. Remote Sens.*, vol. 33, pp. 1609–1617, Jul. 2000.

[4] J. Li, K. Zheng, L. Gao, Z. Han, Z. Li, and J. Chanussot, "Enhanced deep image prior for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504218.

[5] G. M. Espindola, G. Camara, I. A. Reis, L. S. Bins, and A. M. Monteiro, "Parameter selection for region-growing image segmentation algorithms using spatial autocorrelation," *Int. J. Remote Sens.*, vol. 27, no. 14, pp. 3035–3040, Jul. 2006.

[6] C. Ji and H. Tang, "Towards reliable land cover mapping under domain shift: An overview and comprehensive comparative study on uncertainty estimation," *Earth-Sci. Rev.*, vol. 263, Apr. 2025, Art. no. 105070.

[7] X. Lu et al., "Weak-to-strong consistency learning for semisupervised image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510715.

[8] L. Lv and L. Zhang, "Advancing data-efficient exploitation for semi-supervised remote sensing images semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5619213.

[9] X. Qi, Y. Mao, Y. Zhang, Y. Deng, H. Wei, and L. Wang, "PICS: Paradigms integration and contrastive selection for semisupervised remote sensing images semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5602119.

[10] Z. Chen, Y. Hu, and Y. Zhang, "Effects of compression on remote sensing image classification based on fractal analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4577–4590, Jul. 2019.

[11] J. Wei, L. Mi, Y. Hu, J. Ling, Y. Li, and Z. Chen, "Effects of lossy compression on remote sensing image classification based on convolutional sparse coding," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[12] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5622216.

[13] Z. Wang, R. Prabha, T. Huang, J. Wu, and R. Rajagopal, "SkyScript: A large and semantically diverse vision-language dataset for remote sensing," in *Proc. AAAI Conf. Artif. Intell.*, Mar. 2024, vol. 38, no. 6, pp. 5805–5813.

[14] X. Li, C. Wen, Y. Hu, Z. Yuan, and X. X. Zhu, "Vision-language models in remote sensing: Current progress and future trends," *IEEE Geosci. Remote Sens. Mag.*, vol. 12, no. 2, pp. 32–66, Jun. 2024.

[15] M. Bi, M. Wang, Z. Li, and D. Hong, "Vision transformer with contrastive learning for remote sensing image scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 738–749, 2023.

[16] D. Zhang, F. Wang, L. Ning, Z. Zhao, J. Gao, and X. Li, "Integrating SAM with feature interaction for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4513011.

[17] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[18] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[19] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. 18th Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, vol. 9351, Cham, Switzerland. Springer, 2015, pp. 234–241.

[20] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[21] T. Hanyu et al., "AerialFormer: Multi-resolution transformer for aerial image segmentation," *Remote Sens.*, vol. 16, no. 16, p. 2930, Aug. 2024.

[22] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607713.

[23] L. Wang, R. Li, C. Duan, C. Zhang, X. Meng, and S. Fang, "A novel transformer based semantic segmentation scheme for fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[24] R. Li, L. Wang, C. Zhang, C. Duan, and S. Zheng, "A2-FPN for semantic segmentation of fine-resolution remotely sensed images," *Int. J. Remote Sens.*, vol. 43, no. 3, pp. 1131–1155, Feb. 2022.

[25] H. Guo, G. He, W. Jiang, R. Yin, L. Yan, and W. Leng, "A multi-scale water extraction convolutional neural network (MWEN) method for GaoFen-1 remote sensing images," *ISPRS Int. J. Geo-Information*, vol. 9, no. 4, p. 189, Mar. 2020.

[26] B. Cui, W. Jing, L. Huang, Z. Li, and Y. Lu, "SANet: A sea–land segmentation network via adaptive multiscale feature learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 116–126, 2021.

[27] L. Wang, R. Li, D. Wang, C. Duan, T. Wang, and X. Meng, "Transformer meets convolution: A bilateral awareness network for semantic segmentation of very fine resolution urban scene images," *Remote Sens.*, vol. 13, no. 16, p. 3065, Aug. 2021.

[28] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7262–7272.

[29] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501312.

[30] X. Sun et al., "RingMo: A remote sensing foundation model with masked image modeling," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2022, Art. no. 5612822.

[31] K. Cha, J. Seo, and T. Lee, "A billion-scale foundation model for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, early access, May 16, 2024, doi: 10.1109/JSTARS.2024.3401772.

[32] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.

[33] X. Guo et al., "SkySense: A multi-modal remote sensing foundation model towards universal interpretation for Earth observation imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 27662–27673.

[34] N. A. A. Braham, C. M. Albrecht, J. Mairal, J. Chanussot, Y. Wang, and X. X. Zhu, "SpectralEarth: Training hyperspectral foundation models at scale," 2024, *arXiv:2408.08447*.

[35] J. Li et al., "HyperFree: A channel-adaptive and tuning-free foundation model for hyperspectral remote sensing imagery," 2025, *arXiv:2503.21841*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.

[36] K. Kuckreja, M. S. Danish, M. Naseer, A. Das, S. Khan, and F. S. Khan, "GeoChat: Grounded large vision-language model for remote sensing," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2023, pp. 27831–27840.

[37] L. Yang, W. Zhuo, L. Qi, Y. Shi, and Y. Gao, "ST++: Make self-training work better for semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 4268–4277.

[38] J. Wang, C. H. Q. Ding, S. Chen, C. He, and B. Luo, "Semi-supervised remote sensing image semantic segmentation via consistency regularization and average update of pseudo-label," *Remote Sens.*, vol. 12, no. 21, p. 3603, Nov. 2020.

[39] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2022.

[40] G. Li, Z. Liu, and H. Ling, "ICNet: Information conversion network for RGB-D based salient object detection," *IEEE Trans. Image Process.*, vol. 29, pp. 4873–4884, 2020.

[41] J.-X. Wang, S.-B. Chen, C. H. Q. Ding, J. Tang, and B. Luo, "RanPaste: Paste consistency and pseudo label for semisupervised remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 2002916.

[42] G. Camps-Valls, T. V. Bandos Marsheva, and D. Zhou, "Semi-supervised graph-based hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3044–3054, Oct. 2007.

[43] R. M. Haralick and L. G. Shapiro, "Image segmentation techniques," *Comput. Vis. Graph., Image Process.*, vol. 29, no. 1, pp. 100–132, Jan. 1985.

[44] M. Kim, M. Madden, and T. A. Warner, "Forest type mapping using object-specific texture measures from multispectral ikonos imagery," *Photogramm. Eng. Remote Sens.*, vol. 75, no. 7, pp. 819–829, Jul. 2009.

[45] B. Johnson and Z. Xie, "Unsupervised image segmentation evaluation and refinement using a multi-scale approach," *ISPRS J. Photogramm. Remote Sens.*, vol. 66, no. 4, pp. 473–483, Jul. 2011.

[46] J. Radoux and P. Defourny, "Quality assessment of segmentation results devoted to object-based classification," in *Object-Based Image Analysis: Spatial Concepts for Knowledge-Driven Remote Sensing Applications*. Berlin, Germany: Springer, 2008, pp. 257–271.

[47] Y. Xia and Z. Chen, "Quality assessment for remote sensing images: Approaches and applications," in *Proc. IEEE Int. Conf. Syst., Man, Cybern.*, Oct. 2015, pp. 1029–1034.

[48] J. Wang, K. C. Chan, and C. C. Loy, "Exploring clip for assessing the look and feel of images," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, vol. 37, 2023, pp. 2555–2563.

[49] H. Zhou et al., "UniQA: Unified vision-language pre-training for image quality and aesthetic assessment," 2024, *arXiv:2406.01069*.

[50] Z. Tu et al., "MAXIM: Multi-axis MLP for image processing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5769–5780.

[51] R. Li, S. Zheng, C. Zhang, C. Duan, L. Wang, and P. M. Atkinson, "ABCNet: Attentive bilateral contextual network for efficient semantic segmentation of fine-resolution remotely sensed imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 181, pp. 84–98, Nov. 2021.

[52] D. Wang, J. Zhang, B. Du, G.-S. Xia, and D. Tao, "An empirical study of remote sensing pretraining," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5608020.

[53] Y. Lyu, G. Vosselman, G.-S. Xia, A. Yilmaz, and M. Y. Yang, "UAVid: A semantic segmentation dataset for UAV imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 165, pp. 108–119, Jul. 2020.

[54] M. Rahnemoonfar, T. Chowdhury, A. Sarkar, D. Varshney, M. Yari, and R. R. Murphy, "FloodNet: A high resolution aerial imagery dataset for post flood scene understanding," *IEEE Access*, vol. 9, pp. 89644–89654, 2021.

[55] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[56] L. Ma, M. Li, X. Ma, L. Cheng, P. Du, and Y. Liu, "A review of supervised object-based land-cover image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 130, pp. 277–293, Aug. 2017.

[57] S. W. Zamir et al., "iSAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jul. 2019, pp. 28–37.

[58] M. A. Saad, A. C. Bovik, and C. Charrier, "Blind image quality assessment: A natural scene statistics approach in the DCT domain," *IEEE Trans. Image Process.*, vol. 21, no. 8, pp. 3339–3352, Aug. 2012.

[59] A. Mittal, A. K. Moorthy, and A. C. Bovik, "No-reference image quality assessment in the spatial domain," *IEEE Trans. Image Process.*, vol. 21, no. 12, pp. 4695–4708, Dec. 2012.

[60] C. Chen et al., "TOPIQ: A top-down approach from semantics to distortions for image quality assessment," *IEEE Trans. Image Process.*, vol. 33, pp. 2404–2418, 2024.

[61] H. Wang et al., "Blind image quality assessment via adaptive graph attention," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 10, pp. 10299–10309, Oct. 2024.

[62] G. Qin et al., "Data-efficient image quality assessment with attention-panel decoder," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2023, pp. 2091–2100.

[63] S. Yang et al., "MANIQA: Multi-dimension attention network for no-reference image quality assessment," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 1191–1200.

[64] S. Su et al., "Blindly assess image quality in the wild guided by a self-adaptive hyper network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 3667–3676.

[65] D. Muhtar, X. Zhang, P. Xiao, Z. Li, and F. Gu, "CMID: A unified self-supervised learning framework for remote sensing image understanding," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5607817.

[66] F. Bastani, P. Wolters, R. Gupta, J. Ferdinando, and A. Kembhavi, "SatlasPretrain: A large-scale dataset for remote sensing image understanding," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16726–16736.

[67] M. Mendieta, B. Han, X. Shi, Y. Zhu, and C. Chen, "Towards geospatial foundation models via continual pretraining," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16806–16816.

[68] C. J. Reed et al., "Scale-MAE: A scale-aware masked autoencoder for multiscale geospatial representation learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 4088–4099.

[69] W. Cai, K. Jin, J. Hou, C. Guo, L. Wu, and W. Yang, "VDD: Varied drone dataset for semantic segmentation," 2023, *arXiv:2305.13608*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 06:13:00 UTC from IEEE Xplore. Restrictions apply.